

Chapter 8

Quantum Mechanics

8.1 Introduction

In the seventeenth century two rival theories of the nature of light proposed were the wave theory and the corpuscular theory. The Dutch astronomer Huygens (1629–1695) proposed the *wave theory of light* in 1690. He believed that light was a longitudinal wave, and that wave was propagated through a material called the 'ether'. He compared the behaviour of light not with that of water waves but with that of sound. Sound cannot travel through vacuum but light does, and so Huygens proposed that the ether must fill all space (even the vacuum), be transparent and of zero inertia. Clearly a very strange material! However even at the beginning of the twentieth century, scientists were convinced of the existence of the ether.

In 1690, Newton proposed the *corpuscular theory of light*. He believed that light was shot out from a source as small particles and not as wave. He argued that light could not be a wave since we cannot see light when the source is behind an obstacle. He stated that particles of different colours should be of different sizes, the red particles being larger than blue. Since these particles are shot out all the time, according to Newton's theory, the mass of the source of light must reduce and this view was accepted for over a hundred years.

Among the most noteworthy achievements of 19th century science was the realization that light consists of electromagnetic waves. Electromagnetic waves themselves were predicted by James Clerk Maxwell in 1864 on the basis of his theory of electric and magnetic fields and their interrelation. According to this *electromagnetic theory*, light is propagated by varying electric and magnetic fields oscillating at right angles to each other. It travels with a constant velocity of $3 \times 10^8 \text{ ms}^{-1}$ in vacuum. It is unaffected by electric and magnetic fields, and it travels in straight lines in vacuum. It is transverse in nature and hence it may be polarised and it can exhibit interference and diffraction.

8.2 Theory of Black Body Radiation

First let us study how energy is distributed in the spectrum of a black body. The radiation emitted by a black body varies with its temperature. The intensity of radi-

ation corresponding to different wavelengths is measured at different temperatures and plotted as shown in Fig.1. From this study it is observed that

- i) at a given temperature, the energy is not uniformly distributed in the radiation spectrum
- ii) at a given temperature, the intensity of radiation is maximum at a particular wavelength λ_m .
- iii) with increase in temperature, λ_m decreases.
- iv) for all wavelengths, an increase in temperature causes an increase in the energy emission.

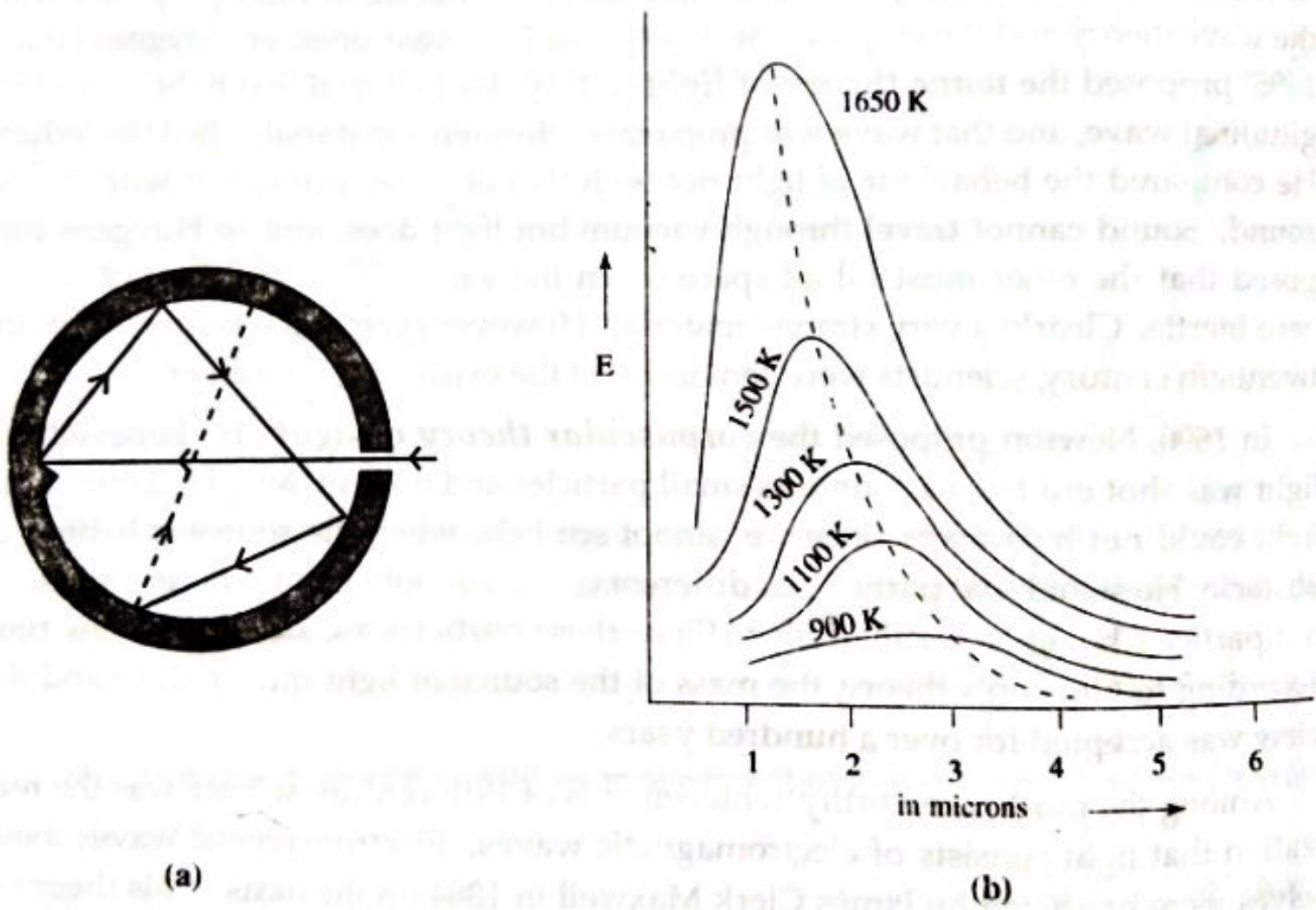


Fig.1 (a) A typical black body and (b) Energy spectrum of a black body

8.2.4 Planck's quantum theory - Black body radiation ✓

In 1901 Planck derived a theoretical expression for the energy distribution on the basis of quantum theory of heat radiation. According to him, energy is emitted in the form of packets or quanta called *photons*. Each photon has an energy $h\nu$ where h is the Planck's constant and ν is the frequency of radiation. According to this theory, the energy is not emitted continuously by the body but it is emitted in certain multiples of the fundamental frequency of the energy emitter (resonator). This means that the energy is emitted as quanta of energies $h\nu, 2h\nu, 3h\nu, \dots$ etc.

8.3 The Photoelectric Effect ✓

The photoelectric effect was discovered in 1887 but was not explained. Einstein in 1905 extended Planck's idea and suggested quantum nature of light and the quantization of energy in his explanation of the photoelectric effect. Einstein received the Nobel prize for physics as his work marked the beginning of quantum theory.

The phenomenon of the emission of electrons from certain metal surfaces when electromagnetic radiation (such as X-rays, ultraviolet rays and in some cases visible light) of sufficient energy falls on them is called Photoelectric effect. The electrons ejected are called photoelectrons.

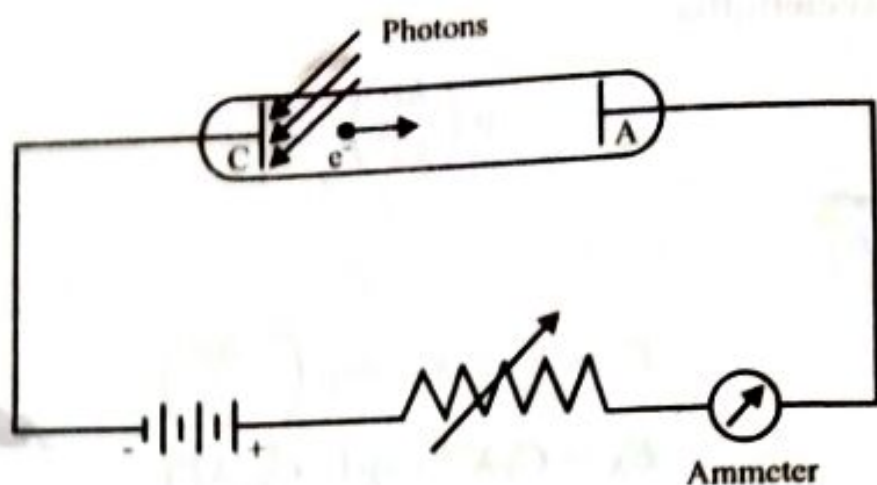


Fig. 2 Schematic of the apparatus for studying the photoelectric effect.

8.3.1 Experimental study

Fig.2 shows the type of apparatus used for studying the photoelectric effect. Two metal (zinc) plates A and C kept inside an evacuated quartz bulb act as anode and cathode respectively. When ultraviolet rays fall on the cathode C (which is connected to the negative terminal of a battery), there is flow of current (as indicated by ammeter). But when light falls on the anode A (which is connected to the positive terminal of the battery) there is no flow of current. From these observations we realise that the particles emitted by the cathode when ultraviolet rays fall on it are nothing but electrons. These emitted electrons are attracted by the anode A causing flow of current. In this experiment photoelectric current is studied as a function of

- i. the intensity of incident light
- ii. the frequency of incident light and
- iii. the potential difference between anode and cathode

8.3.5 Einstein's photoelectric equation ✓

According to Einstein, when light is incident on a metal surface, photons of energy $h\nu$ strike the metal. When a photon transfers its energy $h\nu$ to the energy, it is ejected of the metal.

Let the minimum energy required for emission of electron from the metal surface be $h\nu_0$ where ν_0 is the threshold frequency which depends upon the nature of the metal. This energy is also called the *work function* ϕ , of the metal. When the frequency of the incident light ν_0 is just the threshold value i.e., $\nu = \nu_0$, then the electron is just emitted but with zero velocity. If ν is greater than ν_0 energy $h\nu_0$ is utilised to release the electron. The remaining energy ($h\nu - h\nu_0$) is appearing as kinetic energy of the electron. If m is the mass of electron and ν is its velocity

$$h\nu - h\nu_0 = \frac{1}{2}m\nu_{\max}^2$$
$$\text{i.e., } h\nu - \phi = \frac{1}{2}m\nu_{\max}^2$$

This is *Einstein's photoelectric equation*.

8.4 Matter Waves

With the introduction of the quantum theory, one has to admit a dual nature, wave and particle, for radiant energy. A similar situation arose in 1924, when the French physicist, Louis de Broglie, put forward the bold suggestion that matter, like radiation, has dual nature i.e., matter might exhibit wavelike properties under appropriate conditions.

Louis de Broglie, attempted to develop a theory of radiation in terms of light quanta or photons. He was led to the new conception of matter waves by the following considerations:

- i) Since radiation has been shown to possess a dual nature, wave and particle, matter must also possess dual nature.
- ii) According to the principle of least action in mechanics, a moving particle chooses always that path for which action is minimum. According to the principle of least time in optics, a ray of light always follows a path for which the time of transit is minimum. These suggested that similarity exists between these branches of physics. Hence one can argue that just as radiation has to be supplemented with particle characteristics to explain certain optical phenomena, so also material particles have to be supplemented with a wave aspect for a full understanding of their behaviour.

- iii) According to Bohr, the stable states of electrons in the atom were governed by "integer" rules. In an atom the permitted orbits of the electron are those in which the angular momenta are integer multiples of $h/2\pi$. Since the only phenomena involving integers in physics were those of interference and modes of vibration of stretched strings, both of which imply wave motion, electrons in stable orbits must be also supplemented with wave aspect.

Based on the above concepts, de Broglie made bold suggestion in his doctorate thesis and presented a new idea of matter waves for which he was honoured with the award of the Nobel Prize in 1929. In his thesis he claimed that there should be intimate connection between waves and corpuscles not only in the case of radiation but also in the case of matter. To study the path of a beam of radiation, we use the wave theory, while to calculate the amount of energy of the same beam we use the quantum theory, i.e., we adopt the concept of photon. In a similar manner, as electrons are observed as particle while we calculate the mass, charge and energy must be treated as wave when we find the path of the electrons. When electrons are reflected by an object, we must treat it as though it were a beam of waves.

8.4.1 De Broglie wavelength

The matter waves thus conceived by de Broglie are called de Broglie matter waves. He derived an expression for the wavelength of matter waves on the analogy of radiation.

Based on Planck's theory of radiation, the energy of a photon (quantum) is given by

$$E = h\nu = \frac{hc}{\lambda} \quad (6)$$

where c is the vacuum velocity of light and λ its wavelength. According to Einstein mass-energy relation

$$E = mc^2 \quad (7)$$

From Eqs. (6) and (7),

$$mc^2 = \frac{hc}{\lambda}$$

$$\text{or } \lambda = \frac{h}{mc} = \frac{h}{p} = \frac{h}{\text{momentum}} \quad (8)$$

where $mc = p$ is the momentum associated with photon.

De Broglie suggested that this equation for wavelength is a perfectly general one, applicable to material particles as well as to photons. In the case of particles

$$\text{momentum} = mv.$$

and hence the de Broglie wavelength of a particle is

$$\lambda = \frac{h}{mv} \quad (9)$$

In Eq. (9), the expression for relativistic mass $m_0 \sqrt{1 - (v^2/c^2)}$ has to be used.

If E is the kinetic energy of the material particle, then

$$E = \frac{1}{2}mv^2 = \frac{1}{2} \frac{m^2 v^2}{m} = \frac{1}{2} \frac{p^2}{m}$$

or $p = \sqrt{2mE}$

Hence de Broglie wavelength

$$\lambda = \frac{h}{\sqrt{2mE}} \quad (10)$$

8.4.2 De Broglie wavelength associated with moving objects

To understand the concept of de Broglie waves, let us consider objects of different masses moving with different velocities. Let us compute the de Broglie wavelength of the following.

1. An aeroplane of mass 200 tonnes travelling at a speed of 1000 km/h
2. A 10 g bullet travelling at 500 m/s
3. A smoke particle of mass 10^{-9} gm moving at 1 cm/s
4. An electron moving with a velocity 600 km/s

1. For the aeroplane de Broglie wavelength associated is

$$\begin{aligned} \lambda &= \frac{h}{mv} = \frac{6.626 \times 10^{-34} \times 3600}{(200 \times 1000) \times 1000 \times 10^3} \\ &= \frac{238.536 \times 10^{-32}}{2 \times 10^{11}} \\ &= 119.268 \times 10^{-43} \text{ m} \end{aligned}$$

2. For the bullet

$$\begin{aligned} \lambda &= \frac{6.626 \times 10^{-34}}{10^{-2} \times 500} \\ &= 1.3 \times 10^{-34} \text{ m} \end{aligned}$$

3. For the smoke particle

$$\begin{aligned} \lambda &= \frac{6.626 \times 10^{-34}}{10^{-12} \times 10^{-2}} \\ &= 6.6 \times 10^{-20} \text{ m} \end{aligned}$$

4. For the electron

$$\begin{aligned}\lambda &= \frac{6.626 \times 10^{-34}}{9.1 \times 10^{-31} \times 600 \times 10^3} \\ &= 1.21 \times 10^{-9} \text{ m} \\ &= 1.21 \text{ nm}\end{aligned}$$

In the above calculations we find that the de Broglie wavelength associated with objects/particles we come across in our day-to-day life are far too small because of the smallness of Planck's constant h and larger value of m . There is no experiment that can reveal (or) measure such small wavelengths associated with macroscopic (Laboratory-sized) objects. Hence only for particle of atomic or nuclear size such as electrons, protons, neutrons the wave behavior is observable.

8.4.3 De Broglie wavelength associated with electrons

Let us consider the case of an electron of rest mass m_o and charge e being accelerated by a potential V volts. If v is the velocity attained by the electron due to acceleration then

$$\frac{1}{2}m_o v^2 = eV$$

$$\text{or } v = \sqrt{\frac{2eV}{m}}$$

$$\begin{aligned}\text{de Broglie wavelength } \lambda &= \frac{h}{m_o v} = \frac{h}{m_o} \sqrt{\frac{m_o}{2eV}} \\ &= \frac{h}{\sqrt{2eVm_o}} \\ &= \frac{6.625 \times 10^{-34}}{(2 \times 1.632 \times 10^{-19} \times V \times 9.1 \times 10^{-31})^{1/2}} \\ \lambda &= \frac{12.26}{\sqrt{V}} \text{ \AA} \quad (11)\end{aligned}$$

If the voltage applied to accelerate the electron V is 100 volts then

$$\lambda = \frac{12.26}{\sqrt{100}} = 1.226 \text{ \AA}$$

Since this wavelength corresponds to X-ray region, electrons accelerated by a voltage of 100 volts will exhibit wave nature corresponding to X-ray wavelength. This concept only helped Davisson and Germer to provide experimental evidence on matter waves when they conducted electron diffraction experiments.

8.4.4 Characteristics of matter waves

Since $\lambda = \frac{h}{mv}$,

1. lighter the particle, greater is the wavelength associated with it.
2. lesser the velocity of the particle, longer the wavelength associated with it.
3. for $v = 0$, $\lambda = \infty$, only with moving particle matter wave is associated.
4. whether the particle is charged or not, matter wave is associated with it. This reveals that these waves are not electromagnetic but a new kind of waves.
5. it can be proved that the matter waves travel faster than light.

We know that $E = h\nu$ and

$$E = mc^2$$

$$\therefore h\nu = mc^2 \text{ or}$$

$$\nu = \frac{mc^2}{h}$$

The wave velocity (ω) is given by

$$\omega = \nu\lambda = \frac{mc^2}{h}\lambda = \frac{mc^2}{h} \frac{h}{mv}$$

$$\text{i.e. } \omega = \frac{c^2}{v} \quad (12)$$

As the particle velocity v cannot exceed velocity of light c , ω is greater than velocity of light.

6. No single phenomena exhibits both particle nature and wave nature simultaneously.
7. While position of a particle is confined to a particular location at any time, the matter wave associated with it has some spread as it is a wave. Thus the wave nature of matter introduces an uncertainty in the location of the position of the particle. Heisenberg's uncertainty principle is based on this concept.

8.5 Heisenberg's Uncertainty Principle

As a direct consequence of the dual nature of matter, in 1927, Heisenberg proposed a very interesting principle known as uncertainty principle. If a particle is moving, based on classical mechanics, at any instant we can find its position and momentum. In wave mechanics, we regard a moving particle as a wave group as seen in Fig.9(b). The particle that corresponds to this wave group may be located anywhere within the group at any given time. Of course, in the middle of the group, the probability of finding the particle is more but the probability of finding the particle at any other point inside the wave group is not zero. Narrower the wave group (as in Fig.9(a)) higher will be the accuracy of locating the particle. At the same time, one cannot define the wavelength λ of the wave accurately when the wave group is narrower. Hence measurement of particle's momentum $\left(mv = \frac{h}{\lambda}\right)$ also becomes less accurate. On the other hand, when we consider a wide wave group, wavelength λ can be well defined and hence measurement of momentum becomes more accurate. At the same time, since the width of the wave group is large, locating the position of the particle becomes less accurate. Thus we have the *uncertainty principle*.

"It is impossible to know both the exact position and exact momentum of an object at the same time".

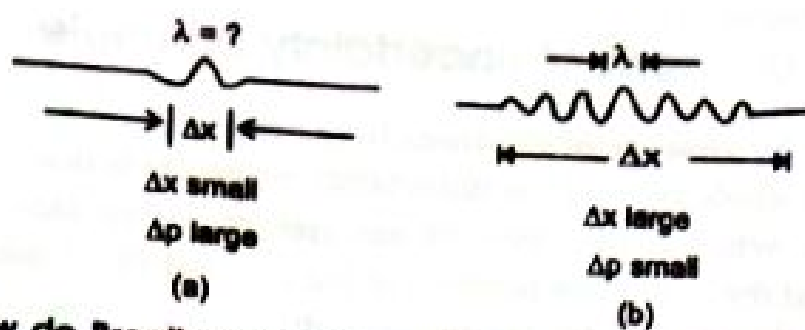


Fig. 9 (a) A narrow de Broglie wave group. Since the position can be precisely determined, Δx is small. As the measurement of λ is less accurate, Δp is large. (b) A wide de Broglie wave group. Measurement of λ is accurate and hence Δp is small whereas, since wave group is wide, Δx is large

If Δx and Δp are the uncertainties in the position and momentum measurements then, according to uncertainty principle.

$$\Delta x \Delta p \geq \frac{h}{4\pi} \quad (13)$$

Thus the uncertainty principle is a direct consequence of the wave nature of particles. Another form of the uncertainty concerns energy and time. In an atomic process let energy E be emitted during the time interval Δt . If the energy is emitted in the form of e.m. waves, we cannot measure the frequency ν of the waves accurately in the limited time available. Let the minimum uncertainty in the number of waves that we count in a wave group be one wave.

$$\text{Since frequency} = \frac{\text{number of wave}}{\text{time interval}}$$

$$\Delta \nu \geq \frac{1}{\Delta t}$$

Hence the corresponding uncertainty in energy ΔE is

$$\Delta E = h \Delta \nu$$

$$\text{and so} \quad \Delta E \geq \frac{h}{\Delta t}$$

$$\text{or} \quad \Delta E \Delta t \geq h$$

A more precise calculation based on the nature of wave groups modifies this result to

$$\Delta E \Delta t \geq \frac{h}{4\pi} \quad (14)$$

This gives uncertainty in the measurement of energy and time of a process.

Thus the more generalized statement of Heisenberg uncertainty principle is

"It is impossible to specify precisely and simultaneously the values of both members of particular pair of physical variables that describe the behavior of an atomic system".

8.5.1 Consequences of uncertainty principle

1. Explanation for absence of electrons in the nucleus

A significant application of the uncertainty principle is finding the answer for the question, whether the electrons are present within atomic nuclei or not? We know that the size of the nuclei is of the order of 10^{-15} meter. If an electron is present within the nuclei, the uncertainty in its position should not exceed 10^{-15} meter. Using the uncertainty relation, the uncertainty in momentum of electron is given by

$$\begin{aligned}\Delta p &\geq \frac{h}{4\pi\Delta x} \\ &\geq \frac{6.62 \times 10^{-34}}{4\pi \times 10^{-15}} \\ &\geq 0.5266 \times 10^{-19} \text{ kg. m/s}\end{aligned}$$

This means that the momentum of the electron in the nucleus must be at least of the order of magnitude $0.5266 \times 10^{-19} \text{ kg. m/s}$

Using the relativistic formula, the energy of the electron E is given by

$$\begin{aligned}E^2 &= p^2 c^2 + m_0^2 c^4 \\ &= (0.5266 \times 10^{-19} \times 3 \times 10^8)^2 \\ &\quad + (9.11 \times 10^{-31} \times (3 \times 10^8)^2)^2 \\ &= (1.5798 \times 10^{-11})^2 + (81.99 \times 10^{-15})^2 \\ &= 2.4958 \times 10^{-22} + 6.722 \times 10^{-27}\end{aligned}$$

$$\text{i.e. } E^2 = 2.4958 \times 10^{-22} \text{ J}$$

$$\text{Hence } E = 1.5798 \times 10^{-11} \text{ J}$$

$$\begin{aligned}&= \frac{1.5798 \times 10^{-11}}{1.6 \times 10^{-19}} \text{ eV} \\ &= 10^8 \text{ eV}\end{aligned}$$

$$E = 100 \text{ MeV}$$

This means that if the electron exists inside the nucleus, it should have the energy of the order of 100 MeV. But the electrons emitted during beta decay of radioactive nuclei have energies less than 5 MeV. Hence, it is evident that electrons do not exist within the nucleus.

2. Proof for existence of protons and neutrons inside nucleus

Extending the calculation of momentum of the electrons for protons and neutrons we get

$$p \simeq 0.5266 \times 10^{-19} \text{ kg m/s.}$$

Hence the kinetic energy K of proton/neutron

$$K = \frac{p^2}{2m_0}$$

For protons and neutrons,

$$m_0 = 1.67 \times 10^{-27} \text{ kg}$$

$$\text{Hence } K = \frac{(0.5266 \times 10^{-19})^2}{2 \times 1.67 \times 10^{-27}} \text{ J}$$

$$= \frac{(0.5266 \times 10^{-19})^2}{2 \times 1.67 \times 10^{-27} \times 1.6 \times 10^{-19}} \text{ eV}$$

$$= 0.05189 \times 10^8 \text{ eV}$$

$$= 5.2 \times 10^6 \text{ eV}$$

$$= 5.2 \text{ MeV}$$

proton

938 neutron 939 .

Since this energy is less than the energies with which these particles are emitted by nuclei, both protons and neutrons can exist inside the nucleus.

8.6 Schrodinger Wave Equation

In solids the carriers of electric current are electrons. We know that the allowed energy levels of an electron belonging to an isolated atom are quantized. Likewise what are the permissible energy levels for an electron free to move within a crystal of specific dimension? Since the free electron can move within the crystal only, it is assumed that the free electron travels in constant potential inside the crystal and is prevented from leaving the crystal by very high energy barriers at the crystal surface. In other words, the electron is trapped in a constant potential well.

Since electrons exhibit wave nature, we can apply wave equation to calculate all possible energy values that a free electron can have inside a crystal.

8.6.1 Schrodinger time dependent wave equation

We know that in classical mechanics undamped, harmonic wave equation has general solution

$$y = A \exp - (iw) \left(t - \frac{x}{v} \right)$$

where y represents the displacement at time ' t ' at a position ' x ', w the angular frequency and v the velocity.

Quantum mechanically we represent the wave associated with a moving particle by a wave function Ψ .

$$\begin{aligned} \Psi(x, t) &= A \exp \left[-iw \left(t - \frac{x}{v} \right) \right] \\ &= A \exp \left[-2\pi i \left(\nu t - \frac{x}{\lambda} \right) \right] \end{aligned} \quad (15)$$

where $w = 2\pi\nu$.

Since $E = h\nu$ and $\lambda = \frac{h}{p}$ the Eq.(15) can be written as

$$\begin{aligned} \Psi(x, t) &= A \exp \left[-2\pi i \left(\frac{Et}{h} - \frac{px}{h} \right) \right] \\ &= A \exp \left[\frac{-2\pi}{h} i (Et - px) \right] \\ &= A \exp \left[\frac{-i}{h} (Et - px) \right] \quad \text{where } h = \frac{h}{2\pi} \end{aligned} \quad (16)$$

As the total energy of the particle is the sum of its K.E. and P.E.,

$$\begin{aligned} E &= \frac{1}{2}mv^2 + V(x) = \frac{1}{2} \frac{p^2}{m} + V(x) \\ \text{i.e., } E\Psi &= \frac{p^2}{2m} \Psi + V\Psi \end{aligned} \quad (17)$$

Differentiating the Eq.(16) with respect to 'x' twice

$$\begin{aligned}\frac{\partial \Psi}{\partial x} &= A \exp \left[\frac{-i}{\hbar} (Et - px) \right] \left(\frac{ip}{\hbar} \right) \\ \frac{\partial^2 \Psi}{\partial x^2} &= A \exp \left[\frac{-i}{\hbar} (Et - px) \right] \left(\frac{i^2 p^2}{\hbar^2} \right) = \frac{-p^2}{\hbar^2} \Psi \\ \text{i.e., } \hbar^2 \frac{\partial^2 \Psi}{\partial x^2} &= -p^2 \Psi\end{aligned}\quad (18)$$

Differentiating Eq.(16) with respect to time 't'

$$\begin{aligned}\frac{\partial \Psi}{\partial t} &= A \exp \left[\frac{-i}{\hbar} (Et - px) \right] \left(\frac{iE}{\hbar} \right) = \frac{-iE\Psi}{\hbar} \\ \text{i.e., } \hbar \frac{\partial \Psi}{i \partial t} &= -E\Psi\end{aligned}\quad (19)$$

Substituting the values of $p^2\Psi$ and $E\Psi$ from Eqs.(18) and (19) in Eq.(17)

$$\begin{aligned}-\hbar \frac{\partial \Psi}{i \partial x} &= \frac{-\hbar^2}{2m} \left[\frac{\partial^2 \Psi}{\partial x^2} \right] + V\Psi \\ \text{i.e., } \frac{\hbar^2}{2m} \left(\frac{\partial^2 \Psi}{\partial x^2} \right) - V\Psi &= \hbar \frac{\partial \Psi}{i \partial t}\end{aligned}\quad (20)$$

This Eq.(20) is known as one dimensional **Schrodinger's time dependent wave equation**. Extending this to three dimensional problem

$$\begin{aligned}\frac{\hbar^2}{2m} \left(\frac{\partial^2 \Psi}{\partial x^2} + \frac{\partial^2 \Psi}{\partial y^2} + \frac{\partial^2 \Psi}{\partial z^2} \right) - V\Psi &= \hbar \frac{\partial \Psi}{i \partial t} \\ \text{i.e., } \frac{\hbar^2}{2m} \nabla^2 \Psi - V\Psi &= -i\hbar \frac{\partial \Psi}{\partial t} \text{ where } \nabla^2 = \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right)\end{aligned}\quad (21)$$

8.6.2 Schrodinger time independent wave equation

This three dimensional Schrodinger equation includes both space and time dependencies. That is, we have $\Psi = \Psi(x, y, z, t)$ and $V = V(x, y, z, t)$. Hence Eq.(20) is referred to as the **time dependent Schrodinger equation**.

For our purpose, it is sufficient to consider situations in which the potential energy of the particle doesn't depend explicitly on time. The forces that act upon it, and hence V , then vary only with the position of the particle. In this case, Eq.(20) reduces to the time independent form. To do so let us rewrite the Eq.(16) as

$$\begin{aligned}\Psi(x, t) &= A \exp \left[\frac{-i}{\hbar} (Et - px) \right] \\ &= A \exp \left(\frac{-iEt}{\hbar} \right) \exp \left(\frac{ipx}{\hbar} \right) \\ &= A\psi(x)\phi(t)\end{aligned}\quad (22)$$

where $\psi(x) = \exp\left(\frac{ipx}{\hbar}\right)$ and $\phi(t) = \exp\left(\frac{-iEt}{\hbar}\right)$

$$\begin{aligned}\therefore \Psi(x, t) &= A\psi \exp \frac{-iEt}{\hbar} \\ \frac{\partial \Psi}{\partial t} &= \frac{-AEi}{\hbar} \psi \exp \frac{-iEt}{\hbar} \\ \text{and } \frac{\partial \Psi}{\partial x} &= A \exp \left(\frac{-iEt}{\hbar} \right) \frac{\partial \psi}{\partial x} \\ \text{or } \frac{\partial^2 \Psi}{\partial x^2} &= A \exp \left(\frac{-iEt}{\hbar} \right) \frac{\partial^2 \psi}{\partial x^2}\end{aligned}$$

Substituting in Eq.(20)

$$\begin{aligned}\frac{\hbar^2}{2m} \cdot A \exp \left(\frac{-iEt}{\hbar} \right) \frac{\partial^2 \psi}{\partial x^2} - V A \psi \exp \left(\frac{-iEt}{\hbar} \right) \\ = \frac{\hbar}{i} \left(\frac{-iAE}{\hbar} \psi \right) \left(\exp \frac{-iEt}{\hbar} \right) \\ \text{i.e., } \frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} - V \psi = -E \psi \\ \text{i.e., } \frac{\partial^2 \psi(x)}{\partial x^2} + \frac{2m}{\hbar^2} (E - V) \psi(x) = 0\end{aligned} \quad (23)$$

where ψ is the time-independent wave function and E is the total energy of the particle. The three dimensional *time independent Schrodinger equation* can be written as

$$\begin{aligned}\left(\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} + \frac{\partial^2 \psi}{\partial z^2} \right) + \frac{2m}{\hbar^2} (E - V) \psi = 0 \\ \text{i.e., } \nabla^2 \psi + \frac{2m}{\hbar^2} (E - V) \psi = 0\end{aligned} \quad (24)$$

Solutions of the Schrodinger equation can be found according to the physical situations.

8.6.3 Physical significance of the wave function ' ψ '

1. The wave function ψ has no direct physical meaning. It is a complex quantity representing the matter wave of electron. It connects the particle nature and its associated wave nature statistically.
2. $\psi\psi^*$ or $|\psi|^2$ is the probability density function. $\psi\psi^* dx dy dz$ gives the probability of finding the electron in the region of space between x and $x + dx$, y and $y + dy$ and z and $z + dz$. If the particle is present

$$\int_{-\infty}^{\infty} \psi\psi^* dx dy dz = 1 \quad (25)$$

3. It can be considered as probability amplitude since it is used to find the location of the particle.